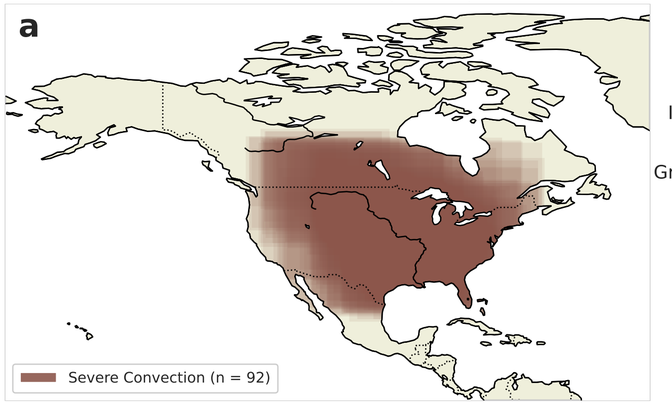
Severe Convection (n = 92)

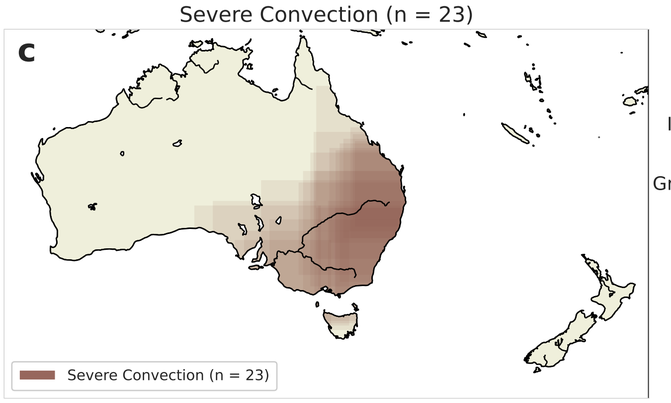


b

	Critical Success Index					False Alarm Ratio					Early Signal					True Positives					False Negatives				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	0.24	0.25	0.23	0.19	0.15	0.42	0.45	0.47	0.55	0.64	1.0	1.0	1.0	1.0	0.9	0.41	0.41	0.35	0.30	0.20	0.54	0.55	0.61	0.65	0.75
GraphCast	0.26	0.26	0.24	0.20	0.16	0.44	0.45	0.46	0.54	0.64	1.0	1.0	1.0	0.9	0.9	0.44	0.42	0.37	0.28	0.23	0.52	0.54	0.59	0.68	0.73
Pangu	0.26	0.27	0.27	0.23	0.18	0.41	0.42	0.43	0.51	0.58	1.0	1.0	1.0	0.9	1.0	0.46	0.46	0.41	0.35	0.28	0.50	0.50	0.55	0.61	0.68
AIFS	0.26	0.26	0.24	0.20	0.17	0.41	0.41	0.45	0.52	0.60	1.0	1.0	1.0	0.9	0.9	0.47	0.42	0.33	0.30	0.23	0.49	0.54	0.63	0.66	0.73

Lead Time [Days]

Severe Convection (n = 23)

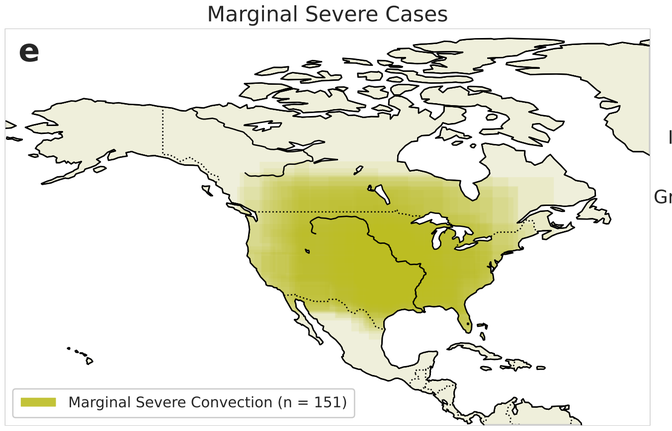


d

	Critical Success Index					False Alarm Ratio					Early Signal					True Positives					False Negatives				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	0.15	0.17	0.14	0.11	0.08	0.66	0.68	0.71	0.68	0.82	0.78	0.83	0.83	0.83	0.57	0.25	0.34	0.34	0.24	0.11	0.71	0.63	0.63	0.72	0.86
GraphCast	0.18	0.17	0.14	0.13	0.12	0.47	0.49	0.52	0.67	0.67	0.94	0.82	0.82	0.76	0.53	0.41	0.43	0.36	0.26	0.22	0.54	0.52	0.60	0.69	0.73
Pangu	0.19	0.19	0.16	0.15	0.09	0.48	0.55	0.63	0.63	0.79	0.94	0.94	0.76	0.88	0.59	0.43	0.43	0.37	0.36	0.19	0.53	0.53	0.58	0.59	0.76
AIFS	0.20	0.19	0.16	0.12	0.11	0.50	0.48	0.46	0.47	0.67	0.94	0.88	0.82	0.82	0.76	0.41	0.36	0.38	0.34	0.16	0.54	0.59	0.57	0.61	0.80

Lead Time [Days]

Marginal Severe Cases



f

	Critical Success Index					False Alarm Ratio					Early Signal					True Positives					False Negatives				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
IFS HRES	0.12	0.12	0.11	0.10	0.08	0.43	0.43	0.47	0.55	0.57	0.73	0.74	0.64	0.57	0.50	0.18	0.18	0.14	0.14	0.10	0.82	0.82	0.86	0.86	0.90
GraphCast	0.11	0.11	0.10	0.08	0.06	0.42	0.43	0.51	0.53	0.61	0.64	0.61	0.55	0.50	0.39	0.16	0.15	0.11	0.09	0.06	0.84	0.85	0.89	0.91	0.94
Pangu	0.11	0.12	0.12	0.10	0.08	0.40	0.39	0.42	0.46	0.57	0.63	0.66	0.62	0.56	0.46	0.17	0.18	0.16	0.12	0.11	0.83	0.82	0.84	0.88	0.89
AIFS	0.11	0.09	0.09	0.07	0.07	0.44	0.42	0.46	0.54	0.60	0.62	0.57	0.57	0.46	0.47	0.18	0.12	0.12	0.08	0.07	0.82	0.88	0.88	0.92	0.93

Lead Time [Days]

← Worse

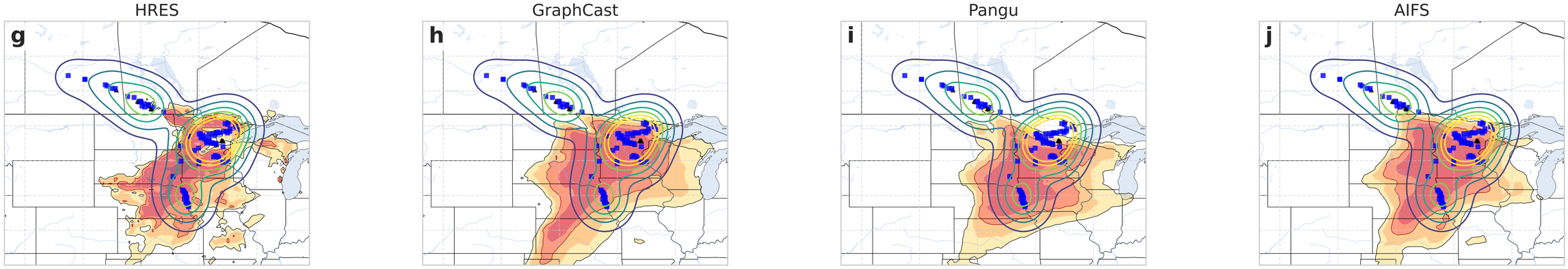
% difference vs IFS HRES

Better →

50 20 10 5 2 1 −1 −2 −5 −10 −20 −50

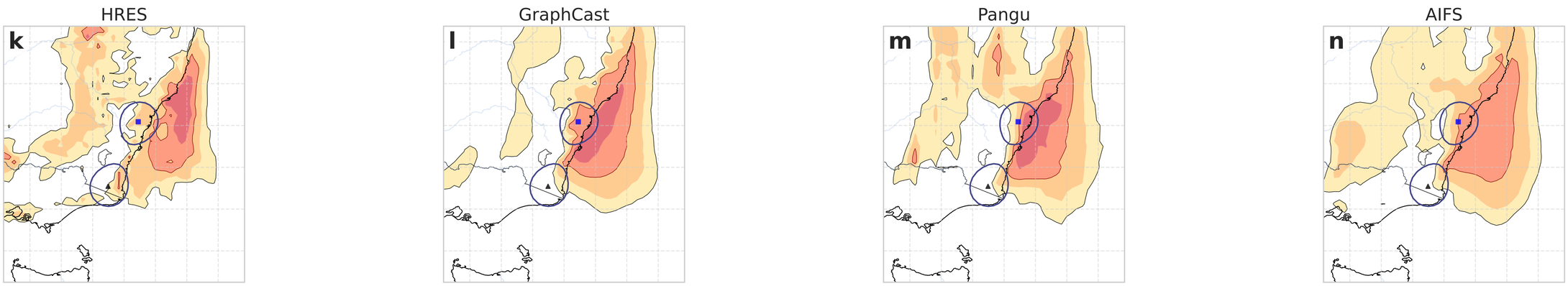
Case 331: Jun 12, 2024 – Jun 13, 2024, Lead Time: 72 h

HRES GraphCast Pangu AIFS



Case 269: Jan 14, 2022 – Jan 15, 2022, Lead Time: 72 h

HRES GraphCast Pangu AIFS



0 10000 15000 22500 30000 50000 75000

Craven-Brooks Significant Severe (m^3/s^3)